

# TSA Pipeline Security Directive — Compliance Report

Cisco Secure Workload | IT-Side Mapping for Natural Gas and Oil  
Pipeline Operators

Cisco Secure Workload — Compliance Mapping Series

Version 1.0 (draft — pending SME review)

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## Cisco Secure Workload — TSA Pipeline Security Compliance Mapping

### How to read this report

This report describes how Cisco Secure Workload (CSW) supports a TSA-designated pipeline owner/operator's compliance posture against the TSA Security Directive Pipeline-2021-02 series cybersecurity requirements on the **IT side** of the IT/OT boundary.

**Scope statement up front.** CSW is a workload-protection and segmentation platform for IT estate — servers, virtual machines, containers, and cloud workloads. CSW does **not** enforce policy on RTUs, flow computers, PLCs, valve actuators, or any other Level 0–2 OT device on the pipeline itself. CSW's TSA SD contribution is at the **IT-side of the IT/OT boundary** and within the IT systems that support pipeline operations — engineering workstations, SCADA jump servers, historian databases, MES/EAM systems, vendor remote-access hosts, identity/PKI services, and the corporate IT systems that touch Critical Cyber Systems on the IT side.

**Companion document.** The accompanying *TSA Pipeline Technical Runbook* walks through deployment, scoping, IT/OT segmentation design, access control, continuous monitoring, vulnerability management on the IT side, the Cybersecurity Incident Response Plan evidence pack, and the Cybersecurity Assessment Plan inputs. Read this report for the audit-narrative view; read the runbook for the deployment view.

**Note on revision letters.** TSA has revised SD Pipeline-2021-02 multiple times (02, 02A, 02B, 02C, 02D, ...). The substantive controls in the III.A–III.D sections have been stable in shape but have been refined in detail. This report is written generically to the current effective revision; reconcile against the active version on the TSA web site before relying on this document in an inspection response.

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## How to read the coverage words

**Full Coverage** on the older reports means Secure Workload can produce that row's artifact. It does not mean the control is met.

**Partial Coverage** means file that export next to another control.

**Evidence Required** means Secure Workload has no record. The customer files the contract, the analysis, or the HR or physical-security evidence.

**Direct, Supporting, and Out of scope** on the newer reports are the same idea in different words. Direct is still evidence, not a pass.

## 1. Executive Summary

The TSA Pipeline Security Directive 2021-02 series imposes mandatory cybersecurity requirements on TSA-designated Critical Pipeline owner/operators in the United States. The directive's substantive cybersecurity content sits in four sections (III.A through III.D) that line up cleanly with the IT-side capabilities of a workload- protection platform: segmentation, access control, continuous monitoring, and reduction of unpatched-system risk.

**What CSW changes for a TSA SD programme:**

- **Section III.A — IT/OT segmentation evidence.** The IT/OT firewall remains your boundary, but CSW enforces deny-by-default *up to* that boundary on the IT side. Undocumented IT pivots toward the IT/OT boundary are structurally removed.
- **Section III.A — Documented data flows.** CSW provides the observed-vs-documented IT → OT flow inventory that lets architecture reviews focus on gaps and changes rather than rebuilding the as-is picture.
- **Section III.B — Access control evidence.** Workload-level least- privilege with quarterly access review and sunset logs for unused rules.
- **Section III.C — Continuous monitoring.** Behavioural detections, conversation-graph anomalies, and per-flow telemetry to your SIEM, paired with your OT-aware sensor stack for end-to-end coverage.
- **Section III.D — Unpatched-system risk reduction.** Continuous CVE inventory per workload with CVSS, exploit availability, and EPSS context; compensating controls registered for delayed patches.
- **CIRP and CAP evidence.** Six-artefact incident reconstruction bundles for your Cybersecurity Incident Response Plan; per-section evidence pack supporting the annual Cybersecurity Assessment Plan self-assessment.

**What CSW does not change.** CSW does not perform Critical Cyber System designation, does not enforce on OT devices, does not perform deep packet inspection of OT protocols (DNP3, IEC 61850, Modbus, OPC-UA), and does not author the Cybersecurity Architecture Design Review or the CIRP. It is the IT-side evidence engine that those governance artefacts depend on.

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## 2. Compliance Posture Summary

The following table summarises CSW's contribution to each in-scope TSA SD requirement. Coverage is rated as:

- **Direct** — CSW is the primary technical control or evidence source for the requirement on the IT side.
- **Supporting** — CSW supplements another control with evidence or enforcement that strengthens the inspection response.
- **Out of scope** — CSW does not address the requirement; relies on other parts of the cyber programme (HR, physical, governance, OT controls).

| SD Section    | Topic                                | Coverage         | CSW contribution                                                                                           |
|---------------|--------------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| Section I     | Applicability + definitions          | Out of scope     | Regulatory determination                                                                                   |
| Section II    | Critical Cyber System identification | Supporting       | Inventory + scope labelling for CSW-managed workloads supporting CCS; cross-reference to your CCS register |
| Section II    | Cybersecurity Coordinator role       | Supporting       | Quarterly evidence pack to the Coordinator                                                                 |
| Section III.A | Network segmentation (IT/OT)         | Direct (IT-side) | Per-site IT-side enclave segmentation; deny-by-default to IT/OT boundary firewall                          |
| Section III.A | Documented data flows                | Direct (IT-side) | Observed IT → OT flow inventory + reconciliation against architecture                                      |
| Section III.B | Access control measures              | Direct (IT-side) | Workload-level least-privilege; quarterly access review with sunset log                                    |
| Section III.C | Continuous monitoring & detection    | Direct (IT-side) | Behavioural rules + flow telemetry to SIEM; conversation-graph anomalies                                   |
| Section III.D | Unpatched-system risk reduction      | Direct (IT-side) | Continuous CVE inventory + compensating-control register for delayed                                       |

| SD Section                               | Topic                                  | Coverage          | CSW contribution                                                               |
|------------------------------------------|----------------------------------------|-------------------|--------------------------------------------------------------------------------|
| Cybersecurity Incident Response Plan     | Plan, test, 24-hour CISA reporting     | Direct (evidence) | patches<br>Six-artefact incident reconstruction bundle                         |
| Cybersecurity Assessment Plan (CAP)      | Annual implementation assessment       | Direct (evidence) | Per-SD-section evidence pack with prior-period delta                           |
| Cybersecurity Architecture Design Review | Periodic architecture review           | Supporting        | Workload inventory + dependency graph + flow inventory as inputs to the review |
| Physical security                        | Physical access controls               | Out of scope      | Facility / PACS process                                                        |
| Personnel & training                     | HR + training                          | Out of scope      | HR / training process                                                          |
| OT device-layer controls                 | RTUs, flow computers, PLCs, IEDs, HMIs | Out of scope      | OT-aware monitoring stack (Cisco Cyber Vision, Claroty, Nozomi, Dragos, etc.)  |

**Posture takeaway.** CSW provides direct IT-side coverage for all four substantive cybersecurity sections (III.A through III.D), the Cybersecurity Incident Response Plan evidence requirement, and the Cybersecurity Assessment Plan self-assessment. It is supporting on Section II inventory and architecture review inputs, and explicitly out of scope for OT-device-layer controls and physical/HR scope. This is consistent with CSW’s positioning as a workload-protection platform: the IT side of the SD is where it earns its place.

### 3. Mapping Table — TSA SD Requirement → CSW Capability → Evidence Artefact

| Requirement                                      | CSW capability                                                                                                   | Evidence artefact (audit-ready)                                                                                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Section II: Critical Cyber System identification | Workload labelling: <code>ccs_id</code> , <code>ccs_function</code> , <code>ot_facing</code> , <code>site</code> | Per-scope inventory CSV with workload identity, OS, owner, CCS reference, OT-facing flag, site, last-seen timestamp |
| Section II: Cybersecurity Coordinator            | Quarterly evidence pack to the Coordinator                                                                       | Standing quarterly report covering §3–§9                                                                            |
| Section III.A: Segmentation policy               | Per-site policy workspace; documented exception list;                                                            | Workspace export per site with                                                                                      |

| Requirement                                | CSW capability                                                              | Evidence artefact (audit-ready)                                           |
|--------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|
|                                            | deny-by-default to boundary firewall                                        | simulation → enforcement transition log                                   |
| Section III.A: IT/OT boundary              | Workload-level deny-by-default for OT-Facing-IT zone                        | Quarterly enforcement diff report                                         |
| Section III.A: Documented data flows       | Observed IT → OT flow inventory + reconciliation against architecture       | IT → OT flow inventory CSV + delta report                                 |
| Section III.A: New-flow change control     | Continuous monitoring for new IT → OT edges                                 | Alert log with disposition column                                         |
| Section III.B: Least privilege             | Workload-level identity-aware policy with rule provenance                   | Policy export with provenance + last-flow timestamp per rule              |
| Section III.B: Cadence-based access review | Quarterly review of allow ruleset; sunset log for stale rules               | Quarterly access review minutes + sunset log                              |
| Section III.C: Behavioural detection       | Out-of-the-box + custom forensic rules tied to CCS function                 | Detection rule catalogue with last-modified dates and SD coverage mapping |
| Section III.C: Anomaly detection           | Conversation-graph anomalies (new edges, port/protocol distribution change) | Anomaly alert log with investigation outcome                              |
| Section III.C: SIEM integration            | Forwarding configuration to Splunk / QRadar / Sentinel / Cisco XDR          | SIEM forwarding configuration + sample alert with full forensic context   |
| Section III.C: Detection latency           | Per-CCS-function MTTD report                                                | Quarterly MTTD trend report                                               |
| Section III.D: CVE inventory               | Per-workload CVE list with CVSS, exploit availability, EPSS                 | CVE export dated to assessment period, prioritised by CCS function        |
| Section III.D: Compensating controls       | Documented in workload workspace notes; enforced via CSW policy             | Compensating-control register for unpatched-but-mitigated workloads       |
| Section III.D: Patch verification          | Package inventory updates within snapshot cycle after patch                 | Diff report showing patched version + automatic CVE drop                  |
| CIRP: Detection and response               | Behavioural rules + quarantine policy + forensic export                     | Quarantine policy applied (timestamped) + six-artefact dossier            |

| Requirement                              | CSW capability                                                  | Evidence artefact (audit-ready)                                                           |
|------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| CIRP: 24-hour CISA reporting             | Standard incident dossier template                              | Per-incident dossier ready to feed CISA submission                                        |
| CAP: Annual self-assessment              | Per-SD-section evidence pack with prior-period delta            | Annual CAP binder organised by SD section + Senior Cybersecurity Officer sign-off summary |
| Cybersecurity Architecture Design Review | Workload inventory + dependency graph + observed flow inventory | Pre-review evidence pack feeding the architecture review                                  |

## 4. Out-of-Scope and Complementary Controls

CSW does **not** address the following SD requirements; reference the noted complementary controls.

| Requirement                                                          | Owner                                          | Complementary control                                               |
|----------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|
| Critical Pipeline designation                                        | TSA                                            | Regulatory determination                                            |
| Critical Cyber System designation                                    | Operations Risk + Senior Cybersecurity Officer | Operational impact analysis                                         |
| Cybersecurity Coordinator appointment                                | Senior Cybersecurity Officer                   | Organisational role assignment                                      |
| Physical security at compressor and pump stations                    | Facilities + Physical Security                 | Site hardening, perimeter, CCTV                                     |
| Personnel & background checks                                        | HR + Senior Cybersecurity Officer              | Background investigation, role-based training                       |
| IT/OT boundary firewall                                              | Network/Firewall team                          | Boundary firewall pair (Cisco Firepower, Palo Alto, Fortinet, etc.) |
| OT device-layer enforcement (RTUs, flow computers, PLCs, IEDs, HMIs) | OT/Operations                                  | OT-aware monitoring (Cyber Vision, Claroty, Nozomi, Dragos)         |
| OT-protocol deep packet inspection (DNP3, IEC 61850, Modbus, OPC-UA) | OT/Operations                                  | OT-aware sensors at the protocol layer                              |
| Physical-to-cyber blast radius modelling                             | Engineering + Operations                       | Process safety + cyber-physical risk analysis                       |
| Pipeline operational continuity planning                             | Operations + Senior Cybersecurity Officer      | Operational continuity plan execution                               |

CSW's role in a complete TSA SD programme is the IT-side segmentation, monitoring, and continuous evidence engine. It explicitly defers the OT device-layer scope to the appropriate specialised stack and the governance / HR / physical scope to the appropriate owners.

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## 5. Suggested Next Steps

For TSA-designated pipeline owner/operators considering CSW within their SD programme:

1. **Confirm CCS inputs.** Validate the Critical Cyber System list and the OT-facing IT tier definition before starting CSW scope design. CSW labels mirror this taxonomy; bad inputs produce bad evidence.
  2. **Pilot on the OT-facing IT tier first.** SCADA jump servers, historian/MES, engineering workstations, and vendor-access hosts give the highest-value wins (Section III.A IT/OT segmentation, Section III.B access control, Section III.C continuous monitoring) with the lowest deployment friction.
  3. **Pair with the existing IT/OT firewall.** CSW does not replace the IT/OT firewall. Position CSW as the IT-side hardening *up to* the firewall, with the firewall remaining the boundary control of record under Section III.A.
  4. **Wire the observed-vs-documented IT → OT flow report into the architecture review cycle.** This is the artefact most likely to surface stale documentation and undocumented operational flows.
  5. **Tie the compensating-control register to the patch programme day one.** The Section III.D requirement is most often missed when patches are delayed for operational reasons; the CSW compensating-control register is the audit-ready evidence that risk has been *reduced*, not just deferred.
  6. **Use the CAP evidence pack to drive the annual self-assessment.** The pack is structured per SD section and produces a prior- period delta automatically, which simplifies the annual review.
  7. **Plan SME review.** This document is a v1 mapping. Before being relied upon in a formal compliance engagement, route to a TSA Pipeline cybersecurity specialist for regulatory accuracy and to a current Cisco Secure Workload SE for product-capability accuracy.
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## 6. Disclaimer

This report and its companion runbook describe how Cisco Secure Workload capabilities can support a TSA Pipeline Security Directive cybersecurity programme on the **IT side** of the IT/OT boundary. They are **not** legal or regulatory advice and do not constitute a finding of compliance.

Specifically:

- TSA Security Directives are issued and revised by the Transportation Security Administration; the active revision letter (02, 02A, 02B, 02C, 02D, ...) and the substantive requirements evolve. Always validate against the **current effective revision**

available from TSA, against your inspector's current guidance, and against any subsequent rulemaking that supersedes the SD.

- Designation as a Critical Pipeline owner/operator is a TSA determination. Identification of Critical Cyber Systems, designation of the Cybersecurity Coordinator, the design of the Cybersecurity Incident Response Plan and the Cybersecurity Assessment Plan, Cybersecurity Architecture Design Review decisions, and incident reporting decisions remain the responsibility of the owner/ operator's Senior Cybersecurity Officer and qualified compliance counsel.
- CSW does not enforce policy on RTUs, flow computers, PLCs, IEDs, HMIs, or any other OT device, and is not certified as an OT- protocol deep-packet-inspection or architecture review tool.

This document should receive subject-matter-expert review for both TSA SD regulatory accuracy and current Cisco product capability before being used in a customer engagement.